HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2013)

Student No.: 15 Name: _____ Marks: / 42

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The solution of $x - \frac{x-1}{2} > 5$ or $1 < x - 11$ is

   A. $x > 9$

   B. $x > 10$

   C. $x > 11$

   D. $x > 12$

(DSE-2013 PII Q05)

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2. Susan sells two cars for $80 080 each. She gains 30% on one and loses 30% on the other. After the transactions, Susan

   A. loses $15 840.

   B. gains $5 544.

   C. gains $10 296.

   D. has no gain and no loss.

(DSE-2013 PII Q10)

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3. The actual area of a playground is $900\,m^2$. Its area on a map is $36\,cm^2$. Then the scale of the map is

A. 1 : 25

B. 1 : 50

C. 1 : 500

D. 1 : 250 000

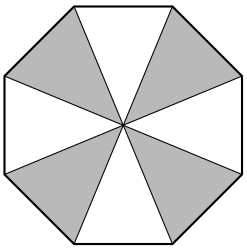
(DSE-2013 PII Q12)

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4. In the figure, the regular octagon is divided into eight identical isosceles triangles and four of them are shaded. The number of axes of reflectional symmetry of the octagon is



A. 2

B. 4

C. 8

D. 16

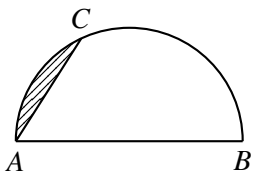
(DSE-2013 PII Q15)

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5. In the figure, the diameter of the semicircle ABC is 3 cm. If chord $AC = 2$ cm, find the area of the shaded region corrected to the nearest $0.01\,cm^2$.



A. $0.23\,cm^2$

B. $0.52\,cm^2$

C. $0.64\,cm^2$

D. $1.07\,cm^2$

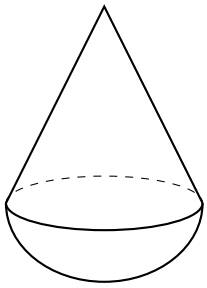
(DSE-2013 PII Q16)

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6. In the figure, the solid consists of a right circular cone and a hemisphere with a common base. The base radius and the height of the circular cone are 3 cm and 4 cm respectively. Find the total surface area of the solid.



A. $30\pi\,\text{cm}^2$

B. $33\pi\,\text{cm}^2$

C. $48\pi\,\text{cm}^2$

D. $51\pi\,\text{cm}^2$

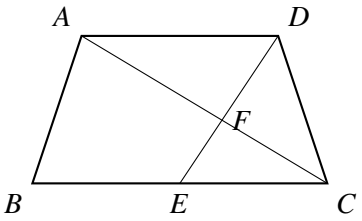
(DSE-2013 PII Q17)

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7. In the figure, $ABCD$ is a trapezium with $AD//BC$, and $AD : BC = 2 : 3$. Let E be the midpoint of BC. AC and DE intersect at F. If the area of ΔCEF is $36\,\text{cm}^2$, then the area of the trapezium $ABCD$ is.



A. $216\,\text{cm}^2$

B. $264\,\text{cm}^2$

C. $280\,\text{cm}^2$

D. $320\,\text{cm}^2$

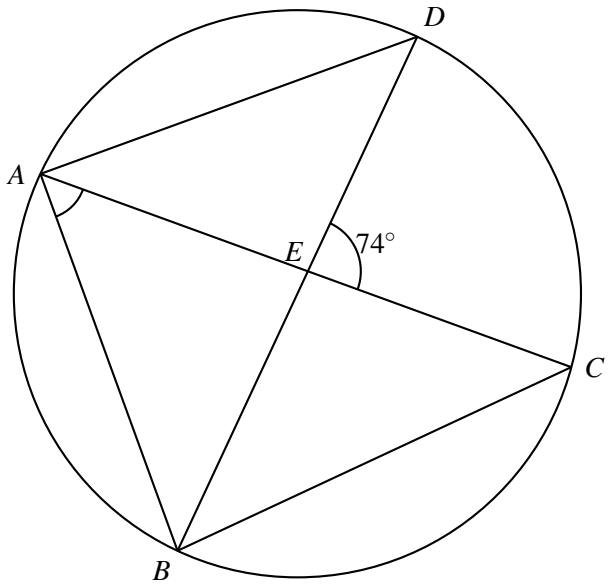
(DSE-2013 PII Q18)

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8. In the figure, $ABCD$ is a circle. AC and BD intersect at E. If $AB = AD$, $AD // BC$, then $\angle BAE =$.



A. 53°

B. 57°

C. 69°

D. 74°

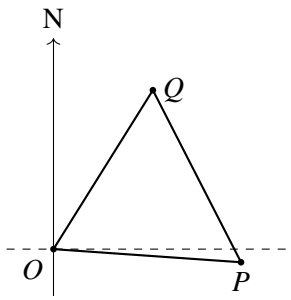
(DSE-2013 PII Q19)

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9. In the figure, the bearing of P from O is $S86^\circ E$ and the bearing of from O is $N32^\circ E$. If P and Q are equidistant from O, then the bearing of P from Q is



A. N24°W

B. N27°W

C. S24°E

D. S27°E

(DSE-2013 PII Q20)

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10. In $\triangle ABC$, $AB : BC : AC = 8 : 15 : 17$. Find $\cos A : \cos C$.

A. 8 : 15

B. 8 : 17

C. 15 : 8

D. 15 : 17

(DSE-2013 PII Q22)

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11. The equation of the circle C is $2x^2+2y^2-4x+8y-5=0$. The coordinates of the points P and Q are $(-1,2)$ and $(4,0)$ respectively. Which of the following is/are true?

I. The radius of C is 5.

II. The midpoint of PQ lies outside C.

III. If G is the centre of C, then $\angle PGQ$ is acute angle.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2013 PII Q25)

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12. Two numbers are randomly drawn at the same time from seven cards numbered 1, 2, 3, 4, 5, 6 and 7 respectively. Find the probability that the product of the numbers drawn is an odd number.

A. $\frac{2}{7}$

B. $\frac{4}{7}$

C. $\frac{12}{49}$

D. $\frac{16}{49}$

(DSE-2013 PII Q26)

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13. If the mean and mode of nine numbers $14, 6, 4, 5, 7, 5, x, y, z$ are 8 and 14 respectively, then the median of these nine number is

A. 5.

B. 6.

C. 7.

D. 8.

(DSE-2013 PII Q27)

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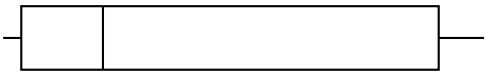
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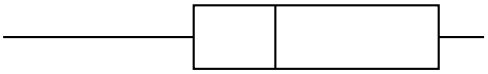
14. The stem-and-leaf diagram shows hourly wages. Which box-and-whisker diagram may represent them?

4	0 2 2 2 4 4 4 7
5	0 0 1 2 2 6 8 9
6	3 5 5 7
7	0
8	2 6
9	5

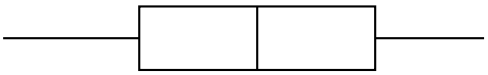
A.



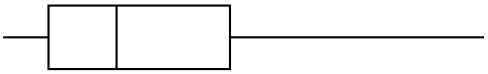
B.



C.



D.



(DSE-2013 PII Q29)

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15. $A00000E00001_{16} =$

A. $10 \times 16^{11} + 14 \times 16^5 + 17$

B. $11 \times 16^{11} + 15 \times 16^5 + 17$

C. $10 \times 16^{12} + 14 \times 16^6 + 272$

D. $11 \times 16^{12} + 15 \times 16^6 + 272$

(DSE-2013 PII Q33)

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16. If $\alpha \neq \beta$ and $\begin{cases} 3\alpha = \alpha^2 - 5 \\ 3\beta = \beta^2 - 5 \end{cases}$, then $\alpha\beta =$

A. 3

B. -3

C. 5

D. -5

(DSE-2013 PII Q35)

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17. The real part of $i+2i^2+3i^3+4i^4$ is

A. 2

B. -2

C. 6

D. -6

(DSE-2013 PII Q36)

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18. The nth term of a sequence is $2n - 19$. Which are true?

I. 25 is a term of the sequence..

II. The sequence has 10 negative terms.

III. The sum of the first n terms of the sequence is $n^2 - 18n$.

A. I only

B. II only

C. I and III only

D. II and III only

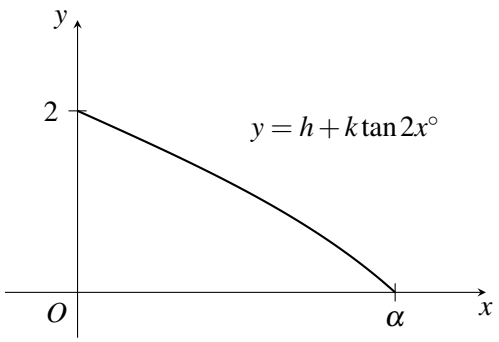
(DSE-2013 PII Q38)

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19. Let h and k be constants. The figure shows the graph of $y = h + k\tan 2x^\circ$, where $0 \le x \le \alpha$. Which of the following are true?



I. $h > 0$

II. $k < 0$

III. $\tan\alpha = \frac{1}{k}$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2013 PII Q39)

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20. If the height of a regular tetrahedron has height 2 cm. then the volume of the tetrahedron is

A. $2\,\text{cm}^3$

B. $\sqrt{3}\,\text{cm}^3$

C. $\sqrt{6}\,\text{cm}^3$

D. $3\sqrt{3}\,\text{cm}^3$

(DSE-2013 PII Q40)

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21. Find the range of k such that the circle $x^2+y^2+2x-2y-7=0$ and the straight line $3x-4y+k=0$ intersect.

A. $-8<k<22$

B. $-8\leq k\leq 22$

C. $k<-22$ or $k>8$

D. $k\leq -22$ or $k\geq 8$

(DSE-2013 PII Q42)

I can finish _____ questions

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